Technical Document #269: Natural Language Processing - Comprehensive Overview

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Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person names, organizations, and locations.

Question answering systems extract answers from a given context. They typically consist of a retriever component and a reader component.

Word embeddings represent words as dense vectors in continuous space. Word2Vec, GloVe, and FastText are popular word embedding techniques.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Key Takeaways:

- The attention mechanism computes a weighted sum of values based on the compatibility between queries and keys.

- Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person nam

- Sentiment analysis determines the emotional tone behind a body of text.